Incident handler's journal

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

Date: October 9, 2025.	**Entry:** 1
Description	A ransomware attack disrupted operations at a small U.S. healthcare clinic after employees received phishing emails. The incident occurred during the *Detection and Analysis* and *Containment, Eradication, and Recovery* phases of the NIST lifecycle, as it involved identifying the infection, isolating affected systems, and restoring services.
Tool(s) used	Email security analysis tools, antivirus software, network monitoring tools (planned for forensic investigation).
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? An organized group of unethical hackers known for targeting healthcare and transportation industries. • **What** happened? Employees received phishing emails containing a malicious attachment. When downloaded, the malware deployed ransomware that encrypted the clinic's critical files. A ransom note demanded payment for the decryption key.

	• **When** did the incident occur? Tuesday morning, approximately 9:00 a.m. • **Where** did the incident happen? At a small healthcare clinic in the United States. • **Why** did the incident happen? The attackers exploited employee vulnerability to phishing emails, gaining network access and deploying ransomware to extort money.
Additional notes	The clinic must focus on cybersecurity awareness training, implement advanced email filtering, and regularly back up data offline. It's also important to review and update the incident response plan to minimize downtime in future incidents.

Date: October 13, 2025.	**Entry:** 2
Description	An employee downloaded a suspicious spreadsheet (.xls) via email that executed a malicious payload. This investigation took place during the *Detection and Analysis* phase, as analysts verified the file's malicious nature through VirusTotal and collected IoCs for further containment.
Tool(s) used	VirusTotal, IDS alerts, Email header analysis, Hashing (SHA256).
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? A threat actor distributing malware through phishing emails. • **What** happened?

	The employee opened a password-protected spreadsheet that executed malware. • **When** did the incident occur? October 13, 2025, between 1:11 p.m. and 1:20 p.m. • **Where** did the incident happen? On an employee's workstation within the company's internal network. • **Why** did the incident happen? The attacker exploited human error and lack of email security awareness to deliver the payload.
Additional notes	The file hash was confirmed as malicious by VirusTotal. Additional IoCs were extracted for network blocking and host scanning. Recommend reinforcing phishing awareness training and enhancing attachment filtering policies.

Date: October 17, 2025	**Entry:** 3
Description	An employee received a phishing email disguised as a job application. This investigation occurred during the *Detection and Analysis* phase, since SOC analysts verified the malware using threat intelligence tools and escalated the case for containment.
Tool(s) used	VirusTotal, Email header analysis, Threat intelligence lookup.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? A malicious actor attempting to distribute malware via phishing email. • **What** happened?

	The employee received an email with a malicious `.exe` attachment that was confirmed as malware. • **When** did the incident occur? July 20, 2022, at 09:30 a.m. • **Where** did the incident happen? On the employee's workstation within the organization's internal mail server. • **Why** did the incident happen? The attacker aimed to compromise the system using social engineering and a malicious executable disguised as a job-related document.
Additional notes	The ticket was escalated to Level-2 SOC analysts for containment and further forensic analysis. Recommend additional employee phishing awareness training and blocking similar sender domains at the email gateway.

Date: October 18, 2025	**Entry:** 4
Description	A web vulnerability allowed an attacker to access customer data from purchase pages. This event spanned the *Containment, Eradication, and Recovery* and *Post-Incident Activity* phases, as the SOC mitigated the exposure, notified customers, and implemented long-term security improvements.
Tool(s) used	Web server log analysis tools, vulnerability scanning tools, incident management platform, and communication with public relations team.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident?

	An external attacker exploiting a web application vulnerability. • **What** happened? The attacker performed a forced browsing attack by modifying order numbers in URLs to access customer purchase confirmation pages and exfiltrate personal data. • **When** did the incident occur? December 28, 2022, 7:20 p.m. PT (reported and investigated between Dec 28–31, 2022). • **Where** did the incident happen? On the company's e-commerce website and web server. • **Why** did the incident happen? Due to missing access control mechanisms and lack of input validation in the web application.
Additional notes	The company collaborated with PR to notify customers and offered free identity protection. Future prevention measures include routine vulnerability scans, penetration testing, URL allowlisting, and stronger authentication for all sensitive pages.

Date: Record the date of the journal entry.	**Entry:** Record the journal entry number.
Description	Provide a brief description about the journal entry.
Tool(s) used	List any cybersecurity tools that were used.

The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? • **What** happened? • **When** did the incident occur? • **Where** did the incident happen? • **Why** did the incident happen?
Additional notes	Include any additional thoughts, questions, or findings.

Date: Record the date of the journal entry.	**Entry:** Record the journal entry number.
Description	Provide a brief description about the journal entry.
Tool(s) used	List any cybersecurity tools that were used.
The 5 W's	Capture the 5 W's of an incident. • **Who** caused the incident? • **What** happened? • **When** did the incident occur? • **Where** did the incident happen? • **Why** did the incident happen?
Additional notes	Include any additional thoughts, questions, or findings.

Need another journal entry template?

If you want to add more journal entries, please copy one of the tables above and paste it into the template to use for future entries.

Reflections/Notes: Were there any specific activities that were challenging for you? Why or why not?

Some of the packet analysis and log correlation activities were challenging at first because they required careful attention to small details and understanding of network behavior. However, with practice, I developed a better ability to identify patterns and interpret indicators of compromise effectively.

Has your understanding of incident detection and response changed since taking this course?

Yes, my understanding has deepened significantly. I now recognize that effective incident response involves not just detection, but also proper documentation, communication, and post-incident analysis to prevent recurrence. The NIST framework provided a structured approach to handle incidents efficiently.

Was there a specific tool or concept that you enjoyed the most? Why?

I enjoyed working with VirusTotal and Suricata, as they allowed me to see how real-world indicators and traffic signatures can reveal malicious behavior. Using these tools made the learning process more practical and closely aligned with real SOC operations.